SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	L.K.PANGAJ	**Reg. No.:**	192511099
Date:	22/07/2026	**Duration:**	60 minutes

Code of Conduct

I _L.K.PANGAJ (192511099)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________PANGAJ_________________

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**

1.1 Explain the Problem in Your Own Word

Cybersecurity threats are increasing rapidly, and many organizations face data breaches because employees are not fully aware of cyber risks. Human errors such as clicking phishing emails, using weak passwords, or sharing confidential information can lead to serious security problems. Traditional cybersecurity training programs are often conducted only once and do not measure employees' understanding regularly. As a result, organizations cannot identify employees who need additional training. The proposed Employee Cyber Security Awareness and Risk Assessment Portal provides continuous cybersecurity education and evaluates employees through online assessments. The system helps improve security awareness, reduce cyber risks, and protect organizational information. Many organizations currently provide cybersecurity awareness through classroom sessions, email guidelines, or occasional online training programs. Employees receive basic security information and complete simple assessments after training. However, these methods are mostly manual and do not continuously monitor employee performance. Organizations often lack a centralized system to evaluate cybersecurity awareness regularly. As a result, identifying employees who require additional training becomes difficult, reducing the effectiveness of existing security programs.

1.2 Identify the Objectives of the Proposed System

The main objective of the proposed system is to improve employees' cybersecurity knowledge through regular awareness training and online assessments. It helps organizations identify employees with higher security risks and provides additional learning resources to improve their knowledge. The system generates reports that help management monitor employee performance and ensure compliance with cybersecurity policies. It also encourages employees to follow safe online practices and reduce security incidents caused by human mistakes. Overall, the system strengthens the organization's cybersecurity culture. The current system helps employees understand basic cybersecurity concepts and improves awareness through training sessions. It is easy to organize and provides general knowledge about cyber threats. However, it has several limitations such as manual record management, limited employee monitoring, and the absence of automated risk assessment. The system cannot continuously track employee performance or provide personalized learning. These limitations reduce the overall effectiveness of cybersecurity awareness programs.

1.3 State the Scope of the Problem

The scope of the proposed system includes employee registration, cybersecurity awareness training, online assessments, risk evaluation, and report generation. Employees can access learning materials, complete quizzes, and view their assessment results through the portal. Administrators can monitor employee progress, manage training content, and generate performance reports. The system can be used in companies, educational institutions, government organizations, and other workplaces. It mainly focuses on improving cybersecurity awareness and reducing cyber risks through continuous learning. The existing system does not provide continuous assessment or identify employees with higher cybersecurity risks. Most organizations conduct awareness training only once or twice a year without monitoring long-term improvement. Manual tracking consumes time and increases administrative work. There is also no automated reporting system to measure employee progress or recommend additional learning. These gaps can be solved by developing an online portal with continuous assessments, risk analysis, and performance monitoring.

2. Identify Key Stakeholders and Challenges

2.1 Identify All the Stakeholders Involved in the System

The main stakeholders of the system include employees, system administrators, IT security teams, company management, HR departments, cybersecurity trainers, and software developers. Employees use the portal to improve their cybersecurity knowledge and complete assessments. Administrators maintain the portal and manage user accounts. IT security teams monitor security risks, while management uses reports for decision-making. HR departments ensure employee participation, and trainers update learning materials. All stakeholders work together to improve organizational cybersecurity. Cybersecurity threats are increasing rapidly, and many organizations face data breaches because employees are not fully aware of cyber risks. Human errors such as clicking phishing emails, using weak passwords, or sharing confidential information can lead to serious security problems. Traditional cybersecurity training programs are often conducted only once and do not measure employees' understanding regularly. As a result, organizations cannot identify employees who need additional training. The proposed Employee Cyber Security Awareness and Risk Assessment Portal provides continuous cybersecurity education and evaluates employees through online assessments

2.2 Explain the Roles and Responsibilities of Each Stakeholder

Employees are responsible for completing cybersecurity training and following secure online practices. System administrators manage users, maintain the portal, and update learning content. IT security teams analyze assessment results and identify high-risk employees. HR departments ensure all employees complete mandatory training programs. Company management reviews reports and implements security policies based on the results. Cybersecurity trainers prepare awareness materials and update them according to new cyber threats. Together, these stakeholders improve employee awareness and reduce security risks. The main objective of the proposed system is to improve employees' cybersecurity knowledge through regular awareness training and online assessments. It helps organizations identify employees with higher security risks and provides additional learning resources to improve their knowledge. The system generates reports that help management monitor employee performance and ensure compliance with cybersecurity policies. It also encourages employees to follow safe online practices and reduce security incidents caused by human mistakes. Overall, the system strengthens the organization's cybersecurity culture

2.3 Discuss the Major Challenges and Issues Faced by the Stakeholders

One of the biggest challenges is the lack of cybersecurity awareness among employees. Many users unknowingly become victims of phishing attacks or use weak passwords, increasing security risks. Another challenge is keeping training content updated as cyber threats change frequently. Organizations must also protect employee assessment data from unauthorized access. Low employee participation and limited cybersecurity knowledge reduce the effectiveness of training programs. Continuous monitoring, regular assessments, and updated learning materials are required to overcome these challenges. The scope of the proposed system includes employee registration, cybersecurity awareness training, online assessments, risk evaluation, and report generation. Employees can access learning materials, complete quizzes, and view their assessment results through the portal. Administrators can monitor employee progress, manage training content, and generate performance reports. The system can be used in companies, educational institutions, government organizations, and other workplaces. It mainly focuses on improving cybersecurity awareness and reducing cyber risks through continuous learning

3. Analyze the Current Approach

3.1 Describe the Existing System or Current Process

Many organizations currently provide cybersecurity awareness through classroom sessions, email guidelines, or occasional online training programs. Employees receive basic security information and complete simple assessments after training. However, these methods are mostly manual and do not continuously monitor employee performance. Organizations often lack a centralized system to evaluate cybersecurity awareness regularly. As a result, identifying employees who require additional training becomes difficult, reducing the effectiveness of existing security programs. Employees are responsible for completing cybersecurity training and following secure online practices. System administrators manage users, maintain the portal, and update learning content. IT security teams analyze assessment results and identify high-risk employees. HR departments ensure all employees complete mandatory training programs. Company management reviews reports and implements security policies based on the results. Cybersecurity trainers prepare awareness materials and update them according to new cyber threats. Together, these stakeholders improve employee awareness and reduce security risks.

3.2 Identify Its Strengths and Limitations

The current system helps employees understand basic cybersecurity concepts and improves awareness through training sessions. It is easy to organize and provides general knowledge about cyber threats. However, it has several limitations such as manual record management, limited employee monitoring, and the absence of automated risk assessment. The system cannot continuously track employee performance or provide personalized learning. These limitations reduce the overall effectiveness of cybersecurity awareness programs. Employees are responsible for completing cybersecurity training and following secure online practices. System administrators manage users, maintain the portal, and update learning content. IT security teams analyze assessment results and identify high-risk employees. HR departments ensure all employees complete mandatory training programs. Company management reviews reports and implements security policies based on the results. Cybersecurity trainers prepare awareness materials and update them according to new cyber threats. Together, these stakeholders improve employee awareness and reduce security risks

3.3 Analyze the Problems or Gaps That Need to Be Addressed

The existing system does not provide continuous assessment or identify employees with higher cybersecurity risks. Most organizations conduct awareness training only once or twice a year without monitoring long-term improvement. Manual tracking consumes time and increases administrative work. There is also no automated reporting system to measure employee progress or recommend additional learning. These gaps can be solved by developing an online portal with continuous assessments, risk analysis, and performance monitoring. One of the biggest challenges is the lack of cybersecurity awareness among employees. Many users unknowingly become victims of phishing attacks or use weak passwords, increasing security risks. Another challenge is keeping training content updated as cyber threats change frequently. Organizations must also protect employee assessment data from unauthorized access. Low employee participation and limited cybersecurity knowledge reduce the effectiveness of training programs. Continuous monitoring, regular assessments, and updated learning materials are required to overcome these challenges.

4. **Propose a Suitable Solution Using Software Engineering Principles**

4.1 Design a Software-Based Solution for the Given Problem

The proposed solution is an Employee Cyber Security Awareness and Risk Assessment Portal developed as a web-based application. Employees can register securely, access cybersecurity learning materials, and complete online quizzes through the portal. The system evaluates employee performance and calculates their cybersecurity risk level based on assessment scores. Administrators can monitor employee progress, generate reports, and update training content whenever required. The portal also sends reminders for pending assessments and recommends additional training for employees who score low. Secure login and encrypted data storage protect employee information from unauthorized access. The system reduces manual work, improves cybersecurity awareness, and helps organizations identify potential security risks before they become serious problems. Overall, the proposed solution creates a secure and effective learning environment for employees. The Agile Software Development Life Cycle (SDLC) model is selected for developing the Employee Cyber Security Awareness and Risk Assessment Portal. Agile divides the project into small development phases called sprints, allowing each module to be developed and tested separately. This approach makes the development process more flexible and efficient. Since cybersecurity threats change frequently, Agile allows the system to be updated regularly with new training materials and features. Continuous feedback from users also helps improve the quality of the software. Therefore, Agile is the most suitable development methodology for this project.

4.2 Describe the Major Functional Modules of the Proposed System

The proposed system consists of several functional modules that work together efficiently. The User Management Module handles employee registration, login, and profile management. The Training Module provides cybersecurity lessons, videos, and learning materials. The Assessment Module conducts quizzes and evaluates employee knowledge. The Risk Assessment Module analyzes assessment scores and identifies employees who require additional training. The Report Module generates detailed reports for management to monitor employee performance. The Notification Module sends reminders about training sessions and assessments. Finally, the Administration Module allows administrators to manage users, update training content, and monitor the overall performance of the portal. Agile is suitable because cybersecurity awareness systems require frequent updates as new cyber threats emerge. Developers can improve the portal by adding new learning materials, assessment questions, and security features without affecting the complete system. Agile supports continuous testing, regular customer feedback, and quick problem-solving throughout development. It also allows changes to be implemented whenever required. Since the project focuses on continuous improvement and employee learning, Agile provides the flexibility needed to keep the system effective and up to date.

4.3 Identify the Functional and Non-Functional Requirements

The functional requirements of the system include employee registration, secure login, access to training materials, online assessments, score calculation, risk evaluation, report generation, and notification services. Employees should be able to view their learning progress and assessment results at any time. The non-functional requirements include security, reliability, scalability, maintainability, and good system performance. The system should provide fast responses, support multiple users simultaneously, and protect confidential employee information through encryption and secure authentication. The interface should also be simple and easy to use for employees with different levels of technical knowledge. The proposed portal follows important software engineering principles to improve system quality. Modularity is achieved by dividing the portal into independent modules such as training, assessment, reporting, and administration. Scalability allows the system to support more employees and additional training courses as the organization grows. Maintainability makes software updates and bug fixes easier without affecting the entire system. Reliability ensures that the portal works continuously and provides accurate assessment results. Usability is achieved through a simple and user-friendly interface that employees can easily understand. Security is implemented using secure login, password encryption, role-based access control, and protected databases to keep employee information safe.

5. **Justify the Chosen Methodology/Framework**

5.1 Select an Appropriate SDLC Model

The Agile Software Development Life Cycle (SDLC) model is selected for developing the Employee Cyber Security Awareness and Risk Assessment Portal. Agile divides the project into small development phases called sprints, allowing each module to be developed and tested separately. This approach makes the development process more flexible and efficient. Since cybersecurity threats change frequently, Agile allows the system to be updated regularly with new training materials and features. Continuous feedback from users also helps improve the quality of the software. Therefore, Agile is the most suitable development methodology for this project. The proposed portal improves employees' cybersecurity awareness through regular learning and online assessments. It helps organizations identify employees who require additional cybersecurity training and reduces the chances of cyberattacks caused by human error. Automated reports make it easier for management to monitor employee performance and improve security policies. The system also saves time by reducing manual record keeping and provides a centralized platform for cybersecurity education. Overall, the portal strengthens organizational security and creates a safer working environment.

5.2 Justify Why the Selected Methodology Is Suitable for the Proposed System

Agile is suitable because cybersecurity awareness systems require frequent updates as new cyber threats emerge. Developers can improve the portal by adding new learning materials, assessment questions, and security features without affecting the complete system. Agile supports continuous testing, regular customer feedback, and quick problem-solving throughout development. It also allows changes to be implemented whenever required. Since the project focuses on continuous improvement and employee learning, Agile provides the flexibility needed to keep the system effective and up to date. The proposed system solves many problems found in traditional cybersecurity training methods. It provides continuous learning instead of one-time awareness sessions and automatically evaluates employee knowledge through online assessments. Employees who score low receive additional training recommendations, helping them improve their cybersecurity skills. Real-time progress tracking reduces administrative work, while detailed reports help management make better decisions. The portal also keeps training materials updated with the latest cybersecurity threats, making employee learning more effective.

5.3 Explain How It Supports Successful Project Development and Maintenance

Agile supports successful project development by encouraging teamwork, regular communication, and continuous testing. Each sprint delivers a working module, making it easier to identify and fix problems at an early stage. User feedback helps improve the portal before the next development cycle begins. After deployment, developers can easily update training materials, improve system performance, and add new features without rebuilding the entire application. This approach reduces maintenance time and ensures that the system remains reliable, secure, and efficient throughout its lifecycle Although the proposed system provides many advantages, it also has some limitations. Developing and maintaining the portal requires financial investment and technical expertise. Some employees may not actively participate in training or complete assessments on time. Internet connectivity issues may affect access to online learning materials. The system also requires regular updates to keep pace with changing cybersecurity threats. Protecting employee assessment data from cyberattacks is another important challenge. Proper maintenance and strong security measures are necessary to ensure the system works effectively.

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6. **Discuss Expected Outcomes and Limitations**

6.1 Describe the Expected Benefits of the Proposed Solution

The proposed portal improves employees' cybersecurity awareness through regular learning and online assessments. It helps organizations identify employees who require additional cybersecurity training and reduces the chances of cyberattacks caused by human error. Automated reports make it easier for management to monitor employee performance and improve security policies. The system also saves time by reducing manual record keeping and provides a centralized platform for cybersecurity education. Overall, the portal strengthens organizational security and creates a safer working environment. Agile supports successful project development by encouraging teamwork, regular communication, and continuous testing. Each sprint delivers a working module, making it easier to identify and fix problems at an early stage. User feedback helps improve the portal before the next development cycle begins. After deployment, developers can easily update training materials, improve system performance, and add new features without rebuilding the entire application. This approach reduces maintenance time and ensures that the system remains reliable, secure, and efficient throughout its lifecycle Although the proposed system provides many advantages, it also has some limitations.

6.2 Explain How the Solution Addresses the Identified Challenges

The proposed system solves many problems found in traditional cybersecurity training methods. It provides continuous learning instead of one-time awareness sessions and automatically evaluates employee knowledge through online assessments. Employees who score low receive additional training recommendations, helping them improve their cybersecurity skills. Real-time progress tracking reduces administrative work, while detailed reports help management make better decisions. The portal also keeps training materials updated with the latest cybersecurity threats, making employee learning more effective. Agile is suitable because cybersecurity awareness systems require frequent updates as new cyber threats emerge. Developers can improve the portal by adding new learning materials, assessment questions, and security features without affecting the complete system. Agile supports continuous testing, regular customer feedback, and quick problem-solving throughout development. It also allows changes to be implemented whenever required. Since the project focuses on continuous improvement and employee learning, Agile provides the flexibility needed to keep the system effective and up to date. The proposed system solves many problems found in traditional cybersecurity training methods.

6.3 Discuss Potential Risks, Implementation Challenges, and Limitations

Although the proposed system provides many advantages, it also has some limitations. Developing and maintaining the portal requires financial investment and technical expertise. Some employees may not actively participate in training or complete assessments on time. Internet connectivity issues may affect access to online learning materials. The system also requires regular updates to keep pace with changing cybersecurity threats. Protecting employee assessment data from cyberattacks is another important challenge. Proper maintenance and strong security measures are necessary to ensure the system works effectively. Agile supports successful project development by encouraging teamwork, regular communication, and continuous testing. Each sprint delivers a working module, making it easier to identify and fix problems at an early stage. User feedback helps improve the portal before the next development cycle begins. After deployment, developers can easily update training materials, improve system performance, and add new features without rebuilding the entire application. This approach reduces maintenance time and ensures that the system remains reliable, secure, and efficient throughout its lifecycle Although the proposed system provides many advantages, it also has some limitations

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/frame work and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks

				organization and clarity.

Criteria	**Mark**
Problem Understanding	/5
Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5